Symmetries in Physics — Exercise Sheet 7

Degeneracy, level splitting, and selection rules

Variant: Questions only

Exercise 1 Restrict each irreducible representation of S_3 to the cyclic subgroup $\mathbb{Z}/3\mathbb{Z}$ of rotations and decompose. Which degeneracies survive the symmetry reduction $S_3 \to \mathbb{Z}/3\mathbb{Z}$ on paper — and which survive in a real quantum system? For the last part, realise $\mathbb{Z}/3\mathbb{Z}$ on the real plane by $R(2\pi/3)$, show that every real commuting matrix is $a\mathbb{1} + bJ$ with $J = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$, and impose symmetry of the Hamiltonian/dynamical matrix.

Exercise 2 Stretch a square into a rectangle: the symmetry drops from D_4 to the Klein subgroup $D_2 = \{e, r^2, s, r^2 s\}$. Build the full correlation table (restriction of each D_4 irreducible to D_2), and show that the doublet E splits into two *distinct* singlets.

Exercise 3 In C_{3v}, with $\chi_E = (2, -1, 0)$ on the classes $(e,\ 2C_3,\ 3\sigma)$: decompose $E \otimes E$ by characters, separate the symmetric and antisymmetric parts, and exhibit explicit invariant combinations in terms of a basis (x, y) of E.

Exercise 4 State the finite-group Wigner–Eckart counting principle: the number of independent reduced matrix elements in $\langle \Gamma' | \, T^{(\Gamma_T)} \, | \Gamma \rangle$ equals the multiplicity of Γ' in $\Gamma_T \otimes \Gamma$. Tabulate this number for all Γ, Γ' in C_{3v} with $\Gamma_T = E$.

Exercise 5 In a C_{3v}-symmetric system the dipole operator has $z \sim A_1$ and $(x, y) \sim E$. Derive the electric-dipole selection rules between levels of symmetries A_1, A_2, E, for axial and planar polarisation separately, and show that $A_1 \leftrightarrow A_2$ is fully forbidden.

Exercise 6 Count the invariants in $E \otimes E \otimes E$ for C_{3v}. What does the answer say about integrals of products of three E-functions — and which Part II computation is this a warm-up for?